# MANOMIT BAL

Autonomy Architecture & Engineering Leadership • Localization • Perception • Sensor Fusion • CV/ML

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## SUMMARY

Autonomy leader and hands-on architect with 10+ years building and shipping production autonomous systems across commercial robotics, aviation safety, and industrial automation. Deep, cross-functional ownership spanning localization and SLAM, perception, computer vision, ML, sensor fusion, and calibration through to full-stack autonomy architecture. Leads cross-functional engineering teams from architecture through field deployment. 15+ patents, and an IEEE standards contributor on autonomy and reliability.

## CORE COMPETENCIES

**Autonomy Architecture** full-stack stack design · modular, hardware-agnostic abstraction · multi-platform reuse · state machines & diagnostics · FMEA / FRACAS

**Localization & SLAM** fault-tolerant state estimation · EKF/ESKF/UKF & adaptive Kalman filtering · GTSAM/iSAM2 factor graphs · 3D LiDAR SLAM · NDT/scan registration · GPS/INS + wheel odometry fusion

**Perception & CV/ML** LiDAR-camera fusion · ground segmentation · object detection (F-RCNN, RF-DETR, TensorRT) · learned segmentation · dynamic object removal · human-aware navigation

**Calibration & Sensing** LiDAR-camera extrinsic calibration · IMU noise characterization · multi-rig calibration pipelines · sensor & compute selection

## EXPERIENCE

**Manager, Fleet Intelligence & Operations** | Gatik AI, Mountain View, CA *Jun 2026 – Present*

- Led fleet intelligence for commercial autonomous middle-mile deployments, connecting operational and fleet-performance data with autonomy, safety, and infrastructure teams to support production reliability and scaling.
- Supported perception and localization reliability work that raised operational runtime ~35%, surfacing false-positive and drift patterns from fleet telemetry to help the autonomy team prioritize fixes.

**Director of Software – Robotics & AI** | Airwai Inc, Los Angeles, CA *Oct 2025 – May 2026*

*Continuing work from the Airwai incubator program at Vebu Inc, Jan 2025 – Sep 2025.*

- Directed a cross-functional engineering team across perception, planning, localization, and infrastructure for an FAA-regulated airfield autonomous vehicle performing debris pickup, pavement monitoring, and perimeter surveillance.
- Owned the full-stack autonomy architecture end to end, from sensing and calibration through localization, perception, and planning.
- Partnered with stakeholders across aviation (FAA), transportation (DOT), and defense and government agencies, delivering three successful pilot deployments across civil and defense airfields that converted into LOIs and continued partnerships.
- Designed a fault-tolerant localization and navigation system fusing LiDAR, GPS/INS, and wheel odometry to hold ~6 cm RMS accuracy in degraded, GPS-challenged environments.
- Built a real-time 3D digital-twin pipeline using SLAM with fused LiDAR, GPS/INS, and camera data, producing georeferenced airfield maps for continuous pavement monitoring and temporal change detection.
- Developed a LiDAR-camera extrinsic calibration pipeline (grid-based) producing validated parameters for real-time sensor fusion across multiple inspection rigs.
- Built AI-driven perception and sensor-fusion algorithms for path planning and situational awareness, deploying TensorRT object-detection pipelines for real-time perception and decision-making.
- Architected a novel autonomous pavement-assessment system from fused LiDAR-camera data, delivering crack dimensions (±3 mm depth via LiDAR ground-plane subtraction, sub-mm width via distance transforms), FOD classification, and marking retroreflectivity estimates aligned with ASTM D6433, FAA AC 150/5210-24, and FAA AC 150/5340-1M, bridging the industry gap between defect detection and measurement.

**Principal Software Engineer** | Vebu Inc, Los Angeles, CA *Sep 2021 – Sep 2025*

- Led architecture and development of autonomy and automation software across commercial lawn-mowing and agricultural robotic platforms, with multiple successful production deployments.
- Drove FMEA and FRACAS processes to harden system reliability and safety across product lines.
- Translated key technical requirements into business milestones, coordinating across sub-teams and mentoring a core software group.
- Owned sensor and compute selection and evaluation for the platform autonomy stack, a core architecture decision spanning perception and localization performance, power, and cost.
- Architected a portable state machine, diagnostics, and ML inference engine reusable across multiple product lines, cutting development time and engineering overhead for new programs. This modular abstraction enabled revalidation across platforms without rebuilding.

***Localization, Perception & Autonomy***

- Overhauled a GPS-based localization system using Gaussian optimization, custom filtering, and a position-correction schema, reducing BOM and development cost by 9%.

- Redesigned a ground-segmentation algorithm with custom filtering to distinguish obstacles in undulating, non-deterministic terrain by fusing 3D LiDAR and an RGBD camera; applied learned segmentation to cut false terrain classifications by 40%.
- Trained and deployed DNNs for feature tracking and matching to estimate robot translation/rotation, and F-RCNNs to counter background clutter and variability.
- Defined a 3D-LiDAR-SLAM strategy to track and recover from faulty pose estimation using lightweight feature NDT matching, and characterized IMU noise parameters to condition data for sensor fusion.
- Migrated UART/RS232 motor control to a robust CANopen architecture, translating embedded-layer metrics into actionable odometry and high-level closed-loop feedback control.

***Automation & AI Systems***

- Led software and sensing for Autocado, a robotic system that cuts, cores, and processes avocados for Chipotle, applying the same perception-and-control autonomy patterns to high-throughput industrial automation.
- Ideated and deployed an ML-based CCTV assessment platform tracking throughput and operational efficiency across 250 QSR locations, yielding a 75% improvement in throughput visibility.
- Developed a teach-and-repeat framework for precise replication of robot-arm trajectories for a robot barista.

**Lead Software Engineer** | Discovery Robotics, Pittsburgh, PA *Apr 2017 – Aug 2021*

- Owned the strategy and design of the entire robot software framework for a commercial cleaning fleet, coordinating across sub-teams and reporting to the CTO.
- Designed a custom 3D localization strategy (3D LiDAR, IMU, wheel encoders) using Gaussian and probabilistic state estimation with fault tolerance, and improved robustness via a custom Kalman-filter sensor-fusion solution with adaptive gains for dead-reckoning conditions.
- Built a real-time 2D/3D SLAM mapping framework with post-processing, map division, and path planning for efficient cleaning patterns, plus AWS-backed map-metadata porting; added a dynamic-object-removal algorithm for drift- and artifact-free 3D maps.
- Improved sensing robustness with PCL-based filtering and integrated a 3D-LiDAR human classifier for human-aware navigation; applied UMBmark to mitigate odometry error on a rear-differential-drive robot.
- Developed a plug-and-play teach-and-repeat algorithm on top of the localization and navigation stack for trajectory learning and replay.

**Robotics Engineering Intern** | Discovery Robotics, Pittsburgh, PA *Summer 2016*

- Developed UWB-based indoor real-time localization with an EKF, designed custom sensor-node PCBs, and contributed to LiDAR-based 3D SLAM and path-planning improvements using PCL and Octomap.

## RESEARCH

**Research Demo: Panoptes, Open-source Multi-Camera Tracking System** *Fall 2016*

N.E. Robotics Colloquium V, Cornell University, Ithaca, NY

- Built Panoptes, an open-source multi-camera 3D tracking system for camera-agnostic rigid-body localization; published at IPSN ’17.

**Research Assistant** | SUNY University at Buffalo, Buffalo, NY *Fall 2015 – Fall 2016*

- Built a visual-odometry pipeline (SIFT/Harris + RANSAC) fused with IMU for CrazyFlie quadcopter tele-operation, and implemented a distributed Markov-based SLAM system with occupancy grids on embedded hardware (Intel Edison, custom Debian kernel, ROS).
- Developed a wearable HCI system extracting real-time gait data from a 48-sensor insole over Bluetooth, visualized in Unity/Oculus Rift with physics-based interaction.
- Benchmarked on-device vs. offloaded computer vision (OpenCV face detection) on Google Glass, quantifying battery, thermal, and frame-rate tradeoffs.

## EDUCATION

**Georgia Institute of Technology** – MS, Computer Science *Dec 2025*

**SUNY University at Buffalo** – BS, Computer Engineering *Dec 2016*

## TECHNICAL SKILLS

**Languages** C, C++, C#, Python, Java

**Autonomy & Perception** ROS 1/2, GTSAM, Ceres, PCL, OpenCV, Eigen, MoveIt, OMPL, Gazebo Ignition

**ML & Inference** TensorFlow, PyTorch, TensorRT, ONNX

**Systems & Tooling** CMake, Git, Docker, AWS, GNU/Linux, CANopen, Modbus TCP/IP, DDS, SQLite, MATLAB, Jira, LaTeX

## PATENTS, PUBLICATIONS & STANDARDS

- **15+ patents** spanning localization, perception, sensor fusion, calibration, and autonomy architecture across three platforms. Google Scholar
- **Publication** M. Bal, J. Yu, K. Dantu, “Panoptes: A Cheap, Extensible, Open-source Multi-Camera Tracking System,” 16th ACM/IEEE IPSN ’17.
- **IEEE Standards Association** Senior Member, Editor, and Voting Member on the Reliability Prediction, Autonomous Agent Alignment, and Robot Task Representation working groups. RAS Senior Member.